



Intelligent Mode Selection for Dual-Functional Radar-Communication in Autonomous Vehicles

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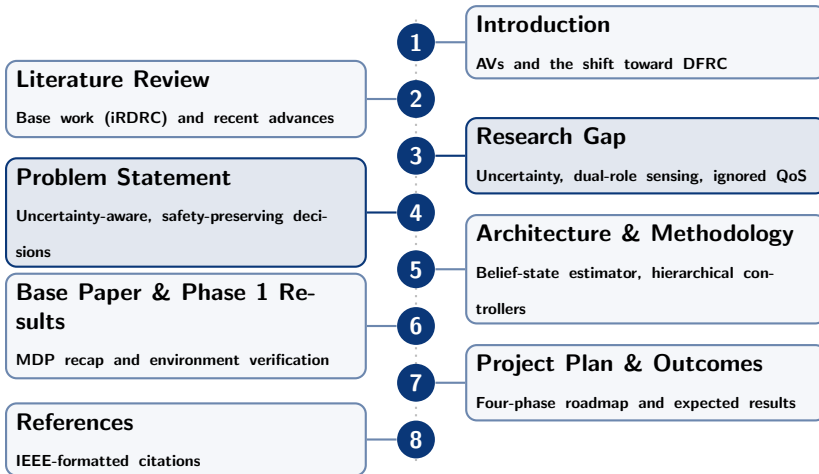
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BTP Review 1

Presentation Outline



Introduction: Why Autonomous Vehicles Need Both Radar and Communication



Autonomous Vehicles (AVs) rely on two distinct but hardware-similar subsystems:

- **Radar (Sensing):** Detects obstacles, pedestrians, and hazards in real time. Critical for collision avoidance, especially in poor visibility (rain, fog, night).
- **Communication (V2X/V2I):** Transmits sensor data, traffic status, and telemetry to edge/cloud infrastructure or nearby vehicles for cooperative driving, route planning, and remote monitoring.

The core tension:

- Both functions increasingly share the same RF front-end, spectrum band, and hardware (mmWave radios also make good radars).
- An AV cannot always do both at once with limited hardware — every time slot spent communicating is a time slot *not* spent sensing, and vice versa.
- This shared-resource conflict motivates **Dual-Functional Radar-Communication (DFRC)** systems.

Evolution: From Separate Systems to Joint Radar-Communication



1. **Separate Systems (Traditional):** Dedicated radar hardware + dedicated communication radio, operating on different spectrum. Simple, but costly, bulky, and spectrum-inefficient as vehicular spectrum demand grows.
2. **Spectrum Congestion Pressure:** Growing V2X traffic and radar density (multiple AVs, mmWave bands) pushed research toward **sharing** hardware/spectrum instead of duplicating it.
3. **Joint Radar-Communication (JRC) / DFRC:** A single hardware platform performs both functions, saving cost, size, and power — the foundation of modern automotive DFRC research.
4. **Fixed / Scheduled Sharing (Early DFRC):** Static time-cycle division or preamble-based reservation (e.g., 802.11ad-style schemes) — radar and comm get a pre-agreed fixed share of time, regardless of actual need.
5. **Adaptive, Learned Sharing (Current Direction):** Since road conditions, traffic, and weather change dynamically, a **learned, adaptive policy** decides mode selection in real time — this is the direction our base paper (iRDRC) takes.

Static/scheduled resource-sharing assumes a predictable world. Reality is not predictable:

- **Weather and road conditions** change continuously — a fixed radar/comm ratio that works on a clear highway is unsafe in fog or on a wet, winding road.
- **Traffic and moving-object density** varies — more radar time is needed near pedestrians/intersections; more comm time is fine on an empty highway.
- **Channel quality** fluctuates — a fixed schedule wastes comm slots on a bad channel and under-uses a good one.
- Fixed schemes cannot trade off **throughput vs. safety** on the fly — they are tuned once, offline, for an average case.

Consequence: The system either wastes capacity (over-cautious radar use) or risks missed hazards (over-aggressive comm use). This motivates replacing fixed rules with a **learned, state-aware decision policy**.

The shift in this line of research:

- Treat mode selection (radar vs. communication) as a **sequential decision-making problem**, not a fixed schedule.
- At every time step, the AV observes its situation — queue backlog, channel quality, road/weather/speed/traffic conditions — and **decides** which mode to use.
- This naturally maps to a **Markov Decision Process (MDP)**, solved using reinforcement learning (e.g., Deep Q-Networks).

This is exactly the approach of our base paper:

iRDRC — An Intelligent Real-Time Dual-Functional Radar-Communication System for Automotive Vehicles (Hieu et al., IEEE Wireless Communications Letters, 2020)

Next: a closer look at this base work, followed by recent papers that build on the same idea.

iRDRC models the AV's radar/communication choice as an **MDP** $\langle S, A, R, P, \gamma \rangle$:

- **State:** (d, c, r, w, v, m) — data queue length, channel quality, and four binary environmental risk factors (road, weather, speed, moving objects); 352 discrete states.
- **Action:** Binary choice each time step — communicate ($a = 0$) or scan with radar ($a = 1$).
- **Reward:** Safety-first design — small positive reward for successful communication, a large penalty (-50) for missing a hazard while communicating, and a risk-scaled reward for successful radar detection.
- **Solver:** Deep Q-Network (DQN) learns the optimal policy $\pi^*(s)$ from experience, without needing a hand-crafted schedule.

Significance: First to frame automotive DFRC mode-selection as a *learned* policy rather than a fixed/static allocation — the direct foundation for this project.

Literature Review: Recent Advances Building on This Idea

Reference	Approach	Modification / Extension over iRDRC-style single-AV DQN
Lee et al. (2022), <i>IEEE TVT</i>	Graph Neural Network (GNN) for JRC	Moves from a single-AV binary decision to a multi-vehicle cooperative setting; needs minimal prior system-model knowledge ; introduces a data-usefulness metric to decide what/whom to transmit to, with a tunable radar-comm performance balance.
Li et al. (2024), <i>IEEE (THz-JSAC)</i>	Dynamic GNN for THz sensing + comm	Extends to Terahertz-band joint sensing-communication across a vehicular network (not one AV); each vehicle can offer sensing-only, comm-only, or both ; a topology-adaptive dynamic GNN outperforms static GNN/optimization baselines by 17–28%.
IEEE TVT (2024)	Hybrid DDPG + DQN for ISAC resource allocation	Replaces the single binary DQN action with a hybrid continuous + discrete action space: DDPG allocates channel/power (continuous), DQN allocates bandwidth (discrete), jointly across many vehicles under spectrum scarcity.

Common trend: All three move from iRDRC's *single-AV, binary-action* MDP toward *multi-vehicle, richer-action* formulations (GNN-based coordination, continuous resource allocation) — pointing to where the open research space likely lies.

The base approach (iRDRC): At every instant, the vehicle must choose between using its sensor (radar) or its communication link — not both at the same time. The decision follows a clear, safety-first rule: prioritize sensing when conditions are risky, and prioritize communication when conditions are safe. This works well for **a single vehicle operating independently**.

Subsequent research extended this idea in three directions:

- Allowing **multiple vehicles to cooperate** and share information.
- Allowing a vehicle to **perform both functions simultaneously**, rather than strictly choosing one.
- Allowing **finer control** over how much attention is given to each function.

Each of these represents genuine progress in flexibility. However, a closer look at this progression reveals an important trade-off.

Four limitations remain consistent across the base work and its extensions:

- **An unrealistic knowledge assumption.** Every method assumes the vehicle already knows its channel quality and risk conditions at each step — in practice this must be estimated from noisy signals.
- **Sensing treated as detection only.** Radar is always modeled purely as a hazard-detection tool; none of these methods recognize that sensing also *resolves the vehicle's own uncertainty*.
- **A diluted safety priority.** As systems became more flexible (cooperative, multi-level), the strict safety-first rule was replaced with a tunable balance — safety became one objective among several, not the dominant priority it was meant to be.
- **Communication constraints are ignored.** Packet deadlines, packet priority, and quality-of-service are absent from every model reviewed — no analytical or learned V2X-ISAC formulation treats communication as anything more than undifferentiated throughput.

The Gap

No existing work models decisions under realistic uncertainty — where sensing is *both* a hazard-detection and uncertainty-reducing action, and communication is deadline- and priority-aware — while preserving a strict, safety-dominant priority instead of a soft trade-off.

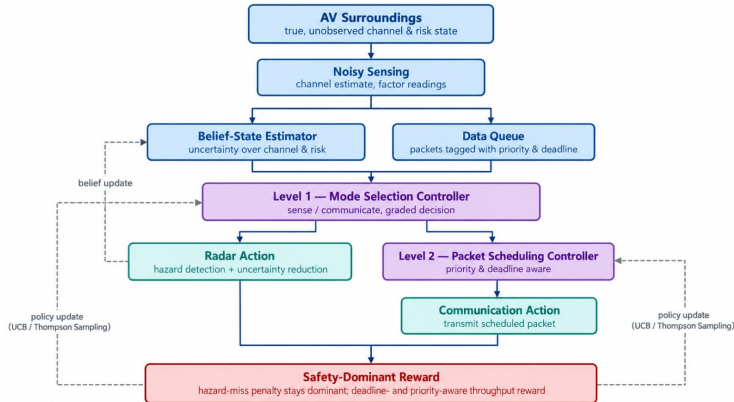
Limitations of Existing Approaches:

- Assume the vehicle already knows its channel and surroundings perfectly
 - In reality, this must be sensed under uncertainty
- Treat radar as all-or-nothing hazard detection only
 - Ignore that sensing can also reduce uncertainty
- Treat every data packet as equally urgent
 - Ignore deadlines, priority, and quality-of-service

Proposed Direction:

- Model the decision under **realistic uncertainty**
- Let sensing serve a **dual role** — detect hazards *and* reduce uncertainty
- Make scheduling **deadline- and priority-aware**
- Keep **safety as the top priority**, always

Proposed System Architecture



Proposed Work: Research Methodology — Architecture Explained



- **AV Surroundings → Noisy Sensing**
 - True channel & risk state is never observed directly, only sensed with noise
- **Belief-State Estimator**
 - Maintains an uncertainty-aware estimate instead of assuming perfect knowledge
- **Data Queue**
 - Packets are now tagged with priority and deadline, not just counted
- **Level 1 — Mode Selection Controller**
 - Decides sense vs. communicate as a graded decision, using belief + queue as context
- **Radar Action**
 - Dual role: detects hazards *and* feeds a belief update back to the estimator
- **Level 2 — Packet Scheduling Controller**
 - When communicating, picks which packet to send based on priority & deadline
- **Safety-Dominant Reward**
 - Hazard-miss penalty stays dominant; trains both controllers via UCB / Thompson Sampling

The MDP tuple: $\langle S, A, R, P \rangle$

- **State space** S (Eq. 2): (d, c, r, w, v, m) , $d \in \{0, \dots, 10\}$, others binary $\rightarrow$ **352 states**
- **Action space** A : $a = 0$ (communicate) or $a = 1$ (radar)
- **Environment model** (Eq. 1): $P(\oplus) = \sum_i [\tau_i p_0^i + (1 - \tau_i) p_1^i]$, $i \in \{r, w, v, m\}$
- **Reward function** (Eq. 3): $+r_1/+r_2$ (comm, no event, good/bad channel); $-r_3$ (comm, event); $+r_4(b+1)$ (radar, event); 0 (radar, no event)
- **Objective** (Eq. 4): $\max_{\pi} G(\pi) = \mathbb{E} \left\{ \sum_t \gamma^t r_{t+1}(\pi) \right\}$

Scope of this review: Implemented in full — state space, action space, environment model, reward function, and objective — everything through the end of Section III. Section IV (the DQN algorithm) is Phase 3.

Sanity checks (MATLAB implementation):

- State space size: **352** (matches Eq. 2 exactly)
- $P(\oplus)$ computed from Eq. 1: **0.1002**

Three fixed policies compared ($T = 500$ steps, no learning yet):

Policy	Total Reward	Discounted G	Mean d	Events
Round-robin	-302.00	-42.19	1.58	48/500
Always-comm	-2044.00	-546.83	0.96	58/500
Always-radar	+595.00	+105.75	9.90	49/500

Inference: Always-radar avoids the $-r_3$ penalty entirely but leaves the queue saturated (near-zero throughput). Always-comm drains the queue but repeatedly eats the -50 penalty. Round-robin sits between the two — exactly the throughput/safety trade-off the reward function is designed to expose, and exactly what a learned policy (Phase 3) must resolve.

Requisites Analysis and Project Plan

Phase	Focus
Phase 1 (<i>Completed</i>)	Part 1 implementation of the base paper: state space (Eq. 2), action space, environment model / unexpected-event probability (Eq. 1), reward function (Eq. 3), and discounted-return objective (Eq. 4) — the full simulation environment, up to but not including the DQN algorithm (Section IV).
Phase 2	Complete the base paper by implementing the DQN algorithm (Section IV) and reproducing the iRDRC baseline results (Section V); then stress-test it across many scenarios — varying channel/event probabilities, reward weights, and environmental conditions — to empirically identify its drawbacks: the perfect-knowledge assumption breaking under estimation noise, the rigidity of the binary action, and the absence of QoS-aware scheduling.
Phase 3	Halfway implementation of the proposed work: belief-state estimator, dual-role radar action, and the Level 1 Mode Selection Controller; evaluate the initial improvement over the base paper's DQN baseline from Phase 2.
Phase 4	Complete the proposed architecture — Level 2 Packet Scheduling Controller, deadline-/priority-aware reward, and full hierarchical training (UCB / Thompson Sampling); run the full comparative evaluation and ablation studies against the base paper and the literature reviewed; consolidate results, write the thesis, and prepare the final defense.

- A validated uncertainty-aware formulation — a belief-state estimator producing a usable channel/risk estimate instead of assuming perfect knowledge, with radar confirmed to serve a **dual role**: hazard detection *and* measurable uncertainty reduction.
- Demonstrated benefit of a **graded, multi-level** mode-selection space over iRDRC's strict binary action.
- A learned hierarchical policy (mode-selection + scheduling controllers) that **outperforms all three Phase 2 baselines** in discounted return $G(\pi)$ — exceeding even always-radar's safety-only ceiling ($G = +105.75$) — while keeping **safety dominant throughout** and maintaining meaningful throughput.
- Reduced miss-detection probability vs. the iRDRC baseline, without sacrificing throughput the way always-radar does; faster convergence than a flat DQN over the same (or richer) state space.
- A working deadline- and priority-aware packet scheduler, with measurably better on-time delivery for high-priority packets than iRDRC's undifferentiated throughput model.
- A direct, quantitative comparison against the base paper and the three recent extensions reviewed, positioning this work's contribution within the current literature.

1. N. Q. Hieu, D. T. Hoang, N. C. Luong, and D. Niyato, "iRDRC: An Intelligent Real-Time Dual-Functional Radar-Communication System for Automotive Vehicles," *IEEE Wireless Communications Letters*, vol. 9, no. 12, pp. 2140–2143, Dec. 2020.
2. J. Lee, Y. Cheng, D. Niyato, Y. L. Guan, and D. González G., "Intelligent Resource Allocation in Joint Radar-Communication With Graph Neural Networks," *IEEE Transactions on Vehicular Technology*, vol. 71, no. 10, pp. 11120–11135, Oct. 2022.
3. X. Li, M. Chen, Y. Hu, Z. Zhang, D. Liu, and S. Mao, "Jointly Optimizing Terahertz Based Sensing and Communications in Vehicular Networks: A Dynamic Graph Neural Network Approach," *IEEE*, 2024.
4. C. Liu, M. Xia, J. Zhao, H. Li, and Y. Gong, "Optimal Resource Allocation for Integrated Sensing and Communications in Internet of Vehicles: A Deep Reinforcement Learning Approach," *IEEE Transactions on Vehicular Technology*, 2024.

Thank You

Questions & Discussion